Stat 122 (Mathematical Statistics II)

Lesson 3.4. Large-sample Confidence Intervals and Sample Size Determination

Learning Objectives

At the end of this lesson, students are expected to construct and interpret confidence intervals for parameters of common distributions based on large samples.

Introduction

The terms **large-sample** and/or **asymptotic** are used to describe confidence intervals that are constructed from asymptotic theory. Of course, the main asymptotic result we have seen so far is the **Central Limit Theorem**. This theorem provides the basis for the large-sample intervals studied in this lesson.

Because these are **large-sample** confidence intervals, this means that the intervals are approximate, so their true confidence levels are *close* to $1-\alpha$ for large sample sizes. We shall present large-sample cofidence intervals for 1. one population mean, μ, 2. one population proportion, p, 3. the difference of two population means,$\mu_1 - \mu_2$, 4. the difference of two population proportions,$p_1 - p_2$.

In each of the situations listed above, we will use a point estimator, say, $\hat{\theta}$, which satisfies

$$Z = \frac{\hat{\theta} - \theta}{\sigma_{\hat{\theta}}} \sim AN(0, 1)$$

for large sample sizes. In this situation, we say that Z is an **asymptotic pivot** because its large-sample distribution is free of all unknown parameters. Because Z follows an approximate standard normal distribution, we can find a value $z_{\alpha/2}$ that satisfies

$$P\left(-z_{\alpha/2} \leq \frac{\hat{\theta} - \theta}{\sigma_{\hat{\theta}}} \leq z_{\alpha/2}\right) \approx 1 - \alpha$$

which, after straightforward algebra (verify!), can be restated as

$$P\left(\hat{\theta} - z_{\alpha/2} \times \sigma_{\hat{\theta}} \leq \theta \leq \hat{\theta} + z_{\alpha/2} \times \sigma_{\hat{\theta}}\right) \approx 1 - \alpha$$

This shows that

$$\hat{\theta} \pm z_{\alpha/2} \times \sigma_{\hat{\theta}}$$

is an approximate $100(1-\alpha)$ percent confidence interval for the parameter θ.

As we have seen in Lesson 3.2, the standard error $\sigma_{\hat{\theta}}$ often depend on unknown parameters (either θ itself or other unknown parameters). This is a problem, because we are trying to compute a confidence interval for θ, and the standard error $\sigma_{\hat{\theta}}$ depends on population parameters which are not known.

The solution is to find a **good** estimator for $\sigma_{\hat{\theta}}$, say, $\hat{\sigma}_{\hat{\theta}}$, then the interval

$$\hat{\theta} \pm z_{\alpha/2} \times \hat{\sigma}_{\hat{\theta}}$$

should remain valid in large samples. The term **good** is used to describe an estimator $\hat{\sigma}_{\hat{\theta}}$ that *approaches* the true standard error $\sigma_{\hat{\theta}}$ (in some sense) as the sample size(s) become(s) large. Thus, we have two approximations at play:

1. the Central Limit Theorem that approximates the true sampling distribution of

$$Z = \frac{\hat{\theta} - \theta}{\sigma_{\hat{\theta}}}$$

2. the approximation arising from using $\hat{\sigma}_{\hat{\theta}}$ as an estimator of $\sigma_{\hat{\theta}}$.

We call $\hat{\sigma}_{\hat{\theta}}$ the **estimated standard error**.

Hence, we will use

$$\hat{\theta} \pm z_{\alpha/2} \times \hat{\sigma}_{\hat{\theta}}$$

as an approximate $100(1-\alpha)$ percent confidence interval for θ. We now present this interval in the context of our four scenarios described earlier. Each of the following intervals is valid for large sample sizes. These intervals may not be valid for small sample sizes.

One population mean

Suppose that $Y_1, Y_2, \cdots, Y_n$ is an *iid* sample with mean μ and variance σ^2 and the problem is estimating μ. In this situation, the Central Limit Theorem says that

$$Z = \frac{\overline{Y} - \mu}{\sigma/\sqrt{n}} \sim AN(0,1)$$

for large n. Here

$$\theta = \mu$$
$$\hat{\theta} = \overline{Y}$$
$$\sigma_{\hat{\theta}} = \frac{\sigma}{\sqrt{n}}$$
$$\hat{\sigma}_{\hat{\theta}} = \frac{S}{\sqrt{n}}$$

where S denotes the sample standard deviation. Thus,

$$\overline{Y} \pm z_{\alpha/2} \times \frac{S}{\sqrt{n}}$$

is an approximate $100(1-\alpha)$ percent confidence interval for the population mean, μ.

Example 3.4.1

The administrators for a hospital would like to estimate the mean number of days required for in-patient treatment of patients between the ages of 25 and 34 years. A random sample of $n = 500$ hospital patients between these ages produced the following sample statistics:

$$\overline{y} = 5.4 \text{ days}$$
$$s = 3.1 \text{ days}$$

Construct a 90% confidence interval for μ, the (population) mean length of stay for this cohort of patients.

SOLUTION:

Here we have $n = 500, \overline{y} = 5.4, s = 3.1,$ and $z_{0.10/2} = 1.645$. Thus, a 90% confidence interval for μ is

$$5.4 \pm 1.645 \times \frac{3.1}{\sqrt{500}} \implies (5.2, 5.6) \text{ days}$$

We are 90% confident that the true mean length of stay, for patients aged 25-34, is between 5.2 and 5.6 days.

REMARK:

We do not (and NEVER!) say the probability that μ is between 5.2 and 5.6 days is 0.90.

One population proportion

Suppose that $Y_1, Y_2, \cdots, Y_n$ is an *iid* Bernoulli(p) sample, where $0 < p < 1$, and that interest lies in estimating the population proportion p. In this situation, the Central Limit Theorem says that

$$Z = \frac{\hat{p} - p}{\sqrt{\frac{p(1-p)}{n}}} \sim AN(0, 1)$$

for large n. Here

$$\begin{aligned} \theta &= p \\ \hat{\theta} &= \hat{p} \\ \sigma_{\hat{\theta}} &= \sqrt{\frac{p(1-p)}{n}} \\ \hat{\sigma}_{\hat{\theta}} &= \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \end{aligned}$$

Thus,

$$\hat{p} \pm z_{\alpha/2} \times \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

is an approximate $100(1-\alpha)$ percent confidence interval for the population proportion, p.

<u>3.4.2</u>

The Women's Interagency HIVS tudy(WIHS) is a large observational study funded by the National Institutes of Health to investigate the effects of HIV infection in women. The WIHS study reports that a total of 1288 HIV-infected women were recruited to examine the prevalence of childhood abuse. Of the 1288 HIV positive women, a total of 399 reported that, in fact, they had been a victim of childhood abuse. Find a 95 percent confidence interval for p, the true proportion of HIV infected women who are victims of childhood abuse.

SOLUTION

Here, $n = 1288$, $z_{0.05/2} = 1.96$, and the sample proportion of HIV childhood abuse victims is

$$\hat{p} = \frac{399}{1288} \approx 0.31$$

Thus, a 95% confidence interval for p is

$$0.31 \pm 1.96 \times \sqrt{\frac{0.31(1-0.31)}{1288}} \implies (0.28, 0.34)$$

We are 95% confident that the true proportion of HIV infected women who are victims of childhood abuse is between 0.28 and 0.34.

Difference of two population means

Let $Y_{11}, Y_{12}, \cdots, Y_{1n_1}$ be an random sample from a $f_Y(y|\mu_1, \sigma_1^2)$, and $Y_{21}, Y_{22}, \cdots, Y_{2n_2}$ be an random sample from a $f_Y(y|\mu_2, \sigma_2^2)$. Assume that $f_Y(y|\mu_1, \sigma_1^2)$ is independent of $f_Y(y|\mu_2, \sigma_2^2)$. Suppose the interest is on estimating the population mean difference $\theta = \mu_1 - \mu_2$.

In this situation, the Central Limit Theorem says that

$$Z = \frac{(\overline{Y}_1 - \overline{Y}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \sim AN(0, 1)$$

for large n_1 and n_2. Here we have

$$\begin{aligned} \theta &= \mu_1 - \mu_2 \\ \hat{\theta} &= \overline{Y}_1 - \overline{Y}_2 \\ \sigma_{\hat{\theta}} &= \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} \\ \hat{\sigma}_{\hat{\theta}} &= \sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}} \end{aligned}$$

where S_1^2 and S_2^2 are the respective sample variances. Thus,

$$(\overline{Y}_1 - \overline{Y}_2) \pm z_{\alpha/2} \times \sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}$$

is an approximate $100(1 - \alpha)$ percent confidence interval for the mean di®erence $\mu_1 - \mu_2$.

Example 3.4.3

A botanist is interested in comparing the growth response of dwarf pea stems to different levels of the hormone indoleaceticacid (IAA). Using 16-day-old pea plants, the botanist obtains 5-millimeter sections and floats these sections on solutions with different hormone concentrations to observe the effect of the hormone on the growth of the pea stem. Let Y_1 and Y_2 denote, respectively, the independent growths that can be attributed to the hormone during the first 26 hours after sectioning for $\frac{1}{2}10^4$ and 10^4 levels of concentration of IAA (measured in mm). Summary statistics from the study are given in following table.

Treatment	Sample size	Sample Mean	SD
$\frac{1}{2}10^4$	$n_1 = 53$	$\bar{y}_1 = 1.03$	$s_1 = 0.49$
10^4	$n_2 = 51$	$\bar{y}_1 = 1.66$	$s_2 = 0.59$

The researcher would like to construct a 99% confidence interval for $\mu_1 - \mu_2$, the mean difference in growths for the two IAA levels.

SOLUTION

Here $z_{0.01/2} = 2.58$, thus, the confidence interval is

$$1.03 - 1.66 \pm 2.58 \times \sqrt{\frac{0.49^2}{53} + \frac{0.59^2}{51}} \implies (-0.90, -0.36)$$

That is, we are 99% confident that the mean difference $\mu_1 - \mu_2$ is between -0.90 and -0.36.

REMARK:

Note that, because this interval does not conclude 0, this analysis suggests that the two (population) means are, in fact, truly different. We shall cover tests of significance in Stat 131 and confidence intervals are useful for this purpose.

Difference of two population proportions

Suppose that we have two independent samples

$$\begin{aligned}&\text{Sample 1: } Y_{11}, Y_{12}, \cdots, Y_{1n_1} \overset{iid}{\sim} Bernoulli(p_1)\\&\text{Sample 2: } Y_{21}, Y_{22}, \cdots, Y_{2n_2} \overset{iid}{\sim} Bernoulli(p_2)\end{aligned}$$

and that interest lies in estimating the population proportion difference $\theta = p_1 - p_2$. In this situation, the Central Limit Theorem says that

$$Z = \frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{\sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}} \sim AN(0, 1)$$

for large n_1 and n_2, where $\hat{p}_1$ and $\hat{p}_2$ are the sample proportions. Here,

$$\begin{aligned}\theta &= p_1 - p_2\\ \hat{\theta} &= \hat{p}_1 - \hat{p}_2\\ \sigma_{\hat{\theta}} &= \sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}\\ \hat{\sigma}_{\hat{\theta}} &= \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}\end{aligned}$$

Thus,

$$(\hat{p}_1 - \hat{p}_2) \pm z_{\alpha/2} \times \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$$

is an approximate $100(1-\alpha)$ percent confidence interval for the difference in the population proportions $p_1 - p_2$.

Example 3.4.4

An experimental type of chicken feed, Ration 1, contains a large amount of an ingredient that enables farmers to raise heavier chickens. However, this feed may be too strong and the mortality rate may be higher than that with the usual feed. One researcher wished to compare the mortality rate of chickens fed Ration 1 with the mortality rate of chickens fed the current best-selling feed, Ration 2. Denote by p_1 and p_2 the population mortality rates (proportions) for Ration 1 and Ration 2, respectively. She would like to get a 95% confidence interval for $p_1 - p_2$. Two hundred chickens were randomly assigned to each ration; of those fed Ration 1, 24 died within one week; of those fed Ration 2, 16 died within one week.

SOLUTION

$$\text{Sample 1} : \hat{p}_1 = \frac{24}{200} = 0.12$$
$$\text{Sample 2} : \hat{p}_2 = \frac{16}{200} = 0.08$$

An approximate 95% confidence interval for the true difference $p_1 - p_2$ is

$$(0.12 - 0.08) \pm 1.96 \times \sqrt{\frac{0.12(1-0.12)}{200} + \frac{0.08(1-0.08)}{200}} \implies (-0.02, 0.10)$$

Thus, we are 95% confident that the true difference in mortality rates is between -0.02 and 0.10. Note that this interval does include 0, so we do not have strong (statistical) evidence that the mortality rates (p_1 and p_2) are truly different.

Sample size determination

In many research investigations, it is of interest to determine how many observations are needed to write a $100(1-\alpha)$ percent confidence interval with a given precision. For example, we might want to construct a 95 percent confidence interval for a population mean in a way so that the confidence interval length is no more than 5 units (e.g. days, inches, dollars, etc.). Sample size determinations is a common in agricultural experiments, clinical trials, engineering investigations, epidemiological studies, etc., and, in most real problems, there is no **free lunch**. Collecting more data is costly! Thus, one must be cognizant not only of the statistical issues associated with sample size determination, but also of the practical issues like cost, time spent in data collection, personnel training, etc.

One population mean

Suppose that $Y_1, Y_2, \cdots, Y_n$ is an iid sample from a $N(\mu, \sigma_0^2)$ population, where σ_0^2 is known. In this situation, an exact $100(1-\alpha)$ percent confidence interval for μ is given by

$$\overline{Y} \pm z_{\alpha/2} \times \frac{\sigma_0}{\sqrt{n}}.$$

Let

$$B = z_{\alpha/2} \times \frac{\sigma_0}{\sqrt{n}}$$

where B denotes the bound on the error in estimation; this bound is called the **margin of error**.

In the setting described above, it is possible to determine the sample size n once we specify these two pieces of information:

- the confidence level, $100(a - \alpha)$
- the margin of error, B

This is true because

$$B = z_{\alpha/2} \times \frac{\sigma_0}{\sqrt{n}} \implies n = \left(\frac{z_{\alpha/2} \times \sigma_0}{B}\right)^2$$

Example 3.4.5

In a biomedical experiment, we would like to estimate the mean remaining life of healthy rats that are given a high dose of a toxic substance. This may be done in an early phase clinical trial by researchers trying to find a maximum tolerable dose for humans. Suppose that we would like to write a 99% confidence interval for μ with a margin of error equal to $B = 2$ days. From past studies, remaining rat lifetimes are well-approximated by a normal distribution with standard deviation $\sigma_0 = 8$ days. How many rats should we use for the experiment?

SOLUTION

Here we have $z_{0.01/2} = 2.58, B = 2,$ and, $\sigma_0 = 8.$ Thus,

$$n = \left(\frac{2.58 \times 8}{2}\right)^2 \approx 107$$

Thus, we would need $n = 107$ rats to achieve these goals.

One population proportion

Suppose that $Y_1, Y_2, \cdots, Y_n$ is an iid $Bernoulli(p)$ sample, where $0 < p < 1$, and that interest lies in writing a confdence interval for p with a prescribed length. In this situation, we know that

$$\hat{p} \pm z_{\alpha/2} \times \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

is an approximate $100(1-\alpha)$ percent confidence interval for p.

To determine the sample size for estimating p with a $100(1-\alpha)$ percent confidence interval, we need to specify the margin of error that we desire,

$$B = z_{\alpha/2} \times \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

We would like to solve this equation for n. However, note that B depends on $\hat{p}$ which, in turn, depends on n. This is a small problem, but we can overcome it by replacing $\hat{p}$ with p^*, a guess for the value of p. Doing this, the last expression becomes

$$B = z_{\alpha/2} \times \sqrt{\frac{p^*(1-p^*)}{n}}$$

and solving this equation for n, we get

$$n = \left(\frac{z_{\alpha/2}}{B}\right)^2 \times p^*(1-p^*)$$

This is the desired sample size to find a $100(1-\alpha)$ percent confidence interval for p with a prescribed margin of error equal to B.

Example 3.4.6

In a Phase II clinical trial, it is posited that the proportion of patients responding to a certain drug is $p^* = 0.4$. To engage in a larger Phase III trial, the researchers would like to know how many patients they should recruit into the study. Their resulting 95% confidence interval for p, the true population proportion of patients responding to the drug, should have a margin of error no greater than $B = 0.03$. What sample size do they need for the Phase III trial?

SOLUTION

Here, we have $B = 0.03, p^* = 0.4,$ and $z_{0.05/2} = 1.96$. The desired sample size is

$$n = \left(\frac{1.96}{0.03}\right)^2 \times 0.4(1-0.4) \approx 1024$$

Thus, their Phase III trial should recruit around 1024 patients.

REMARK

If there is no sensible guess for p available, use $p^* = 0.5$. In this situation, the resulting value for n will be as large as possible. Put another way, using $p^* = 0.5$ gives the most conservative solution (i.e., the largest sample size, n). This is true because

$$n = \left(\frac{z_{\alpha/2}}{B}\right)^2 \times p^*(1-p^*)$$

when viewed as a function of p^*, is maximized when $p^* = 0.5$.